

SUPPORTING INFORMATION

Morphology of four common and phylogenetically distant ant species varies along disturbance and aridity gradients in the Caatinga dry forest

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Table S1. Global multimeric disturbance index (GMDI), water deficit (WD), presence of ant species, and mean (M) and variance (V) of traits HBN, north-eastern Brazil. CL = ratio of clypeus length to mesosoma length, ML = ratio of mandible length to mesosoma length, WL = mesosoma or Weber's length, IOW = ratio of inter-ocular width to mesosoma length, EL = ratio of eye length to mesosoma length, P = count of hairs crossing mesosoma profile, LEG = ratio of leg length (combined length of femur and tibia) to mesosoma length and AL = ratio of antenna (combined funicle and scape) length to mesosoma length.

Ant species	Pl ot	GMDI	DH	Traits															
				CL		ML		WL		IOW		EL		P		LEG		AL	
				M	V	M	V	M	V	M	V	M	V	M	V	M	V	M	V
<i>Dinoponera quadriceps</i>	P0	52.86		0.26	0.0000	0.54	0.0000	8.93	0.0083	0.43	0.0001	0.16	0.0000	10.40		1.73	0.0010	1.75	0.0013
	2	8	952	1	1	1	3	3	3	3	4	4	8	0	4.30000	9	9	0	4
	P0	21.95	107	0.27	0.0000	0.56	0.0002	8.83	0.1296	0.43	0.0000	0.16	0.0000	10.25		1.74	0.0020	1.70	0.0028
	7	8	7	8	2	2	2	3	3	0	9	5	2	0	4.91667	2	3	9	8
	P1	36.71		0.27	0.0000	0.56	0.0000	8.66	0.0277	0.44	0.0000	0.16	0.0000			1.75	0.0003	1.74	0.0006
	0	4	926	7	4	2	9	7	8	2	9	1	6	9.800	6.70000	8	3	3	5
	P1	48.27		0.28	0.0000	0.57	0.0000	8.72	0.0648	0.44	0.0002	0.15	0.0000			1.74	0.0003	1.72	0.0000
	1	1	929	0	5	3	2	2	1	6	6	9	4	9.000	3.00000	5	0	6	2
	P1	11.00	105	0.27	0.0000	0.57	0.0000	8.83	0.0277	0.45	0.0000	0.16	0.0000			1.78	0.0000	1.76	0.0000
	4	8	1	7	5	2	3	3	8	3	7	3	7	9.333	4.33333	6	6	1	6
	P1	20.37	108	0.28	0.0001	0.56	0.0001	8.53	0.0750	0.43	0.0000	0.15	0.0000	12.40	20.8000	1.77	0.0022	1.80	0.0018
	5	9	6	9	0	3	4	3	0	8	9	6	2	0	0	0	9	1	5
	P1			0.25	0.0000	0.53	0.0000	9.03	0.0472	0.43	0.0000	0.15	0.0000	12.40	36.3000	1.75	0.0009	1.75	0.0006
	7	2.743	658	5	1	1	3	3	2	9	2	1	9	0	0	3	9	0	5
	P2			0.26	0.0000	0.55	0.0002	8.66	0.0416	0.43	0.0001	0.15	0.0000	15.20	11.2000	1.74	0.0005	1.76	0.0009
	0	9.071	931	9	4	0	6	7	7	5	4	4	1	0	0	6	4	6	9
	P2	22.43		0.26	0.0000	0.56	0.0001	8.90	0.0638	0.44	0.0001	0.14	0.0000			1.74	0.0003	1.72	
	1	8	734	6	1	6	3	0	9	6	4	6	8	9.600	2.30000	6	1	7	03.
	P2	45.76		0.26	0.0001	0.56	0.0000	8.94	0.1481	0.43	0.0000	0.15	0.0002			1.72	0.0004	1.72	0.0032
	3	6	802	7	2	5	1	4	5	5	1	5	3	9.667	2.33333	0	8	2	4
	P2	31.25		0.28	0.0000	0.56	0.0003	8.75	0.1759	0.43	0.0000	0.16	0.0000	10.75		1.73	0.0009	1.73	0.0015
	6	3	947	1	3	7	4	0	3	3	1	2	6	0	8.91667	9	9	9	4
	P2	26.84		0.25	0.0000	0.53	0.0002	9.00	0.0555	0.43	0.0000	0.15	0.0001		15.8000	1.74	0.0007	1.72	0.0017
	7	5	664	9	5	7	0	0	6	7	1	6	2	9.400	0	1	2	7	2
	P2	26.16		0.27	0.0000	0.55	0.0001	8.60	0.0638	0.43	0.0000	0.15	0.0001	11.40	20.3000	1.75	0.0009	1.79	0.0019
	9	1	833	2	6	8	7	0	9	4	5	5	5	0	0	3	0	9	0
	P3	58.06		0.28	0.0001	0.55	0.0002	8.66	0.0833	0.42	0.0000	0.15	0.0000	13.20	10.7000	1.70	0.0032	1.76	0.0015
	0	6	667	1	0	8	0	7	3	7	4	4	2	0	0	9	5	6	8
<i>Ectatomma muticum</i>	P0	52.86		0.16	0.0000	0.43	0.0000	3.41	0.0253	0.46	0.0003	0.17	0.0001			1.48	0.0010	1.43	0.0020
	2	8	952	4	3	0	6	3	3	5	2	1	9	3.200	0.20000	8	5	5	2
	P0	22.31	994	0.22	0.0000	0.42	0.0001	3.48	0.0208	0.46	0.0000	0.14	0.0000	2.600	0.80000	1.49	0.0015	1.45	0.0021

	4	6		4	1	9	7	0	9	3	3	7	3			5	4	0	0
	P0	21.95	107	0.22	0.0002	0.43	0.0006	3.28	0.1320	0.46	0.0000	0.13	0.0000			1.53	0.0010	1.47	
	7	8	7	7	0	5	6	0	0	3	9	7	9	2.800	0.70000	5	1	0	
	P1	36.71		0.22	0.0000	0.42	0.0000	3.42	0.0057	0.44	0.0000	0.14	0.0000			1.51	0.0007	1.44	0.0004
	0	4	926	2	1	8	2	7	8	3	5	0	1	2.200	0.20000	4	2	4	9
	P1	48.27		0.19	0.0000	0.43	0.0001	3.48	0.0453	0.46	0.0001	0.15	0.0000			1.48	0.0020	1.45	0.0026
	1	1	929	8	9	9	2	0	3	5	4	4	3	1.600	0.30000	1	6	4	2
	P1	11.00	105	0.24	0.0002	0.42	0.0000	3.12	0.0142	0.44	0.0000	0.14	0.0000			1.51	0.0009	1.53	0.0034
	4	8	1	6	1	9	7	0	2	6	3	3	4	2.000	0.50000	7	6	1	0
	P1	36.37	105	0.19	0.0000	0.45	0.0003	3.44	0.0302	0.48	0.0003	0.15	0.0001			1.51	0.0030	1.48	0.0016
	6	1	1	8	4	1	2	0	2	5	2	8	2	1.800	0.20000	4	4	4	5
	P1			0.16	0.0000	0.44	0.0001	3.36	0.0368	0.46	0.0000	0.14	0.0001			1.50	0.0001	1.47	0.0002
	7	2.743	658	5	3	1	2	0	9	1	4	7	4	3.200	1.70000	4	3	6	3
	P2	45.76		0.21	0.0000	0.43	0.0001	3.33	0.0266	0.46	0.0000	0.15	0.0000			1.47	0.0006	1.45	0.0025
	3	6	802	2	7	8	0	3	7	2	6	1	8	3.000	0.50000	7	7	0	3
	P2	19.23	100	0.20	0.0000	0.44	0.0003	3.45	0.0275	0.45	0.0004	0.16	0.0000			1.47	0.0013	1.46	0.0029
	5	1	3	7	2	1	6	3	6	5	5	0	3	3.400	1.80000	6	7	1	3
	P2	31.25		0.23	0.0003	0.43	0.0005	3.12	0.3708	0.45	0.0002	0.15	0.0000			1.50	0.0029	1.52	0.0094
	6	3	947	3	6	6	0	7	1	8	5	1	8	2.200	1.06667	9	3	0	5
	P2	26.84		0.19	0.0000	0.43	0.0003	3.61	0.0231	0.47	0.0002	0.15	0.0000			1.49	0.0026	1.43	0.0031
	7	5	664	2	3	2	5	3	1	3	5	7	5	2.200	1.20000	6	0	0	3
	P2	26.16		0.22	0.0001	0.42	0.0002	3.37	0.0213	0.45	0.0002	0.14	0.0000			1.47	0.0031	1.46	0.0018
	9	1	833	1	2	5	9	3	3	7	9	7	4	3.200	0.20000	1	5	8	4
	P0	52.86		0.16	0.0000	0.42	0.0016	0.84	0.0025	0.59	0.0000	0.18	0.0003			1.66	0.0019	2.24	0.0085
	2	8	952	5	7	5	0	7	6	8	8	1	4	1.600	1.30000	9	0	0	9
	P0	22.31		0.18	0.0002	0.44	0.0000	0.83	0.0016	0.60	0.0008	0.17	0.0002			1.71	0.0030	2.25	
	4	6	994	4	8	8	9	3	7	8	8	6	7	3.200	0.70000	1	6	8	0
	P0	16.87	101	0.18	0.0000	0.43	0.0000	0.90	0.0011	0.62	0.0004	0.18	0.0000			1.71	0.0016	2.11	0.0053
	8	6	4	5	5	0	8	0	1	2	6	5	5	0.600	0.80000	2	8	9	6
	P1	20.37	108	0.19	0.0003	0.46	0.0001	0.81	0.0014	0.57	0.0006	0.19	0.0002			1.71	0.0008	2.28	0.0043
	5	9	6	7	7	7	2	3	4	4	4	7	1	1.600	0.80000	3	0	1	6
	P1	36.37	105	0.18	0.0001	0.43	0.0003	0.86	0.0018	0.60	0.0002	0.18	0.0003			1.65	0.0010	2.17	
	6	1	1	6	5	5	9	0	9	5	1	6	0	1.400	0.30000	8	9	3	3
	P2			0.17	0.0003	0.42	0.0001	0.89	0.0007	0.60	0.0009	0.18	0.0000			1.70	0.0026	2.17	0.0004
	0	9.071	931	2	8	5	8	3	8	4	4	7	3	2.000	0.50000	1	3	2	8
	P2	22.43		0.19	0.0000	0.42	0.0003	0.88	0.0008	0.60	0.0007	0.19	0.0000			1.74	0.0040	2.19	0.0068
	1	8	734	0	4	4	1	0	9	6	3	0	4	2.200	0.70000	3	7	0	6
	P2		103	0.18	0.0003	0.45	0.0001	0.87	0.0007	0.59	0.0001	0.17	0.0004			1.71	0.0007	2.19	0.0022
	2	9.146	7	3	1	0	1	3	8	5	9	6	1	1.800	0.70000	8	5	9	6
	P2	45.76		0.18	0.0002	0.43	0.0009	0.86	0.0005	0.62	0.0010	0.19	0.0000			1.76	0.0014	2.24	0.0032
	3	6	802	4	0	9	2	7	6	4	9	2	3	3.200	1.20000	2	0	7	8
	P2	19.23	100	0.18	0.0002	0.46	0.0002	0.80	0.0030	0.57	0.0012	0.17	0.0002			1.70	0.0039	2.25	0.0049
	5	1	3	2	4	3	8	7	0	0	8	4	0	1.400	0.80000	4	7	0	5
	P2	26.84		0.18	0.0000	0.43	0.0006	0.90	0.0007	0.62	0.0001	0.18	0.0000			1.77	0.0057	2.22	0.0027
	7	5	664	4	3	4	7	7	8	5	9	4	3	3.000	0.50000	3	3	2	8
	P2	26.60		0.19	0.0000	0.43	0.0004	0.86	0.0005	0.61	0.0010	0.19	0.0000			1.73	0.0020	2.25	0.0114
	8	6	789	2	3	1	2	7	6	6	2	2	3	3.200	1.70000	9	3	6	1
	P3	58.06		0.18	0.0002	0.42	0.0006	0.85	0.0025	0.60	0.0012	0.18	0.0002			1.80	0.0028	2.29	0.0144
	0	6	667	0	8	9	0	3	6	2	1	8	1	2.800	2.70000	5	3	3	4
<i>Dorymyrmex</i>	P0	22.31	994	0.15	0.0000	0.35	0.0007	1.30	0.0007	0.27	0.0003	0.16	0.0001	0.000	0.00000	1.89	0.0030	1.91	0.0010

	4	6		3	1	2	0	7	8	5	8	3	2			3	0	9	8
	P0	21.95	107	0.14	0.0000	0.35	0.0000	1.22	0.0042	0.27	0.0000	0.16	0.0000			1.89	0.0025	1.98	0.0162
	7	8	7	2	2	5	7	0	2	3	8	9	1	0.000	0.00000	1	5	3	9
	P0	16.87	101	0.13	0.0000	0.37	0.0005	1.25	0.0053	0.28	0.0001	0.17	0.0003			1.92	0.0063	1.97	0.0041
	8	6	4	3	6	3	2	3	3	8	4	6	3	0.000	0.00000	3	0	6	9
	P1	36.71		0.13	0.0002	0.36	0.0005	1.19	0.0035	0.27	0.0002	0.17	0.0000			1.83	0.0015	1.93	0.0026
	0	4	926	9	2	3	7	3	6	4	3	3	1	0.000	0.00000	1	9	8	4
	P1	11.00	105	0.12	0.0000	0.35	0.0000	1.20	0.0141	0.27	0.0002	0.16	0.0001			1.92	0.0018	2.05	0.0104
	4	8	1	1	2	9	4	7	1	1	0	6	5	0.000	0.00000	0	7	6	2
<i>thoracicus</i>	P1	20.37	108	0.13	0.0002	0.36	0.0001	1.12	0.0018	0.27	0.0001	0.16	0.0003			1.88	0.0114	2.04	0.0101
	5	9	6	0	0	1	8	7	9	2	4	6	4	0.000	0.00000	9	1	9	6
	P1	36.37	105	0.12	0.0000	0.36	0.0001	1.31	0.0008	0.26	0.0002	0.15	0.0002			1.85	0.0113	1.95	0.0051
	6	1	1	7	1	0	2	3	9	9	5	8	2	0.000	0.00000	4	9	0	7
	P2		103	0.12	0.0000	0.36	0.0002	1.31	0.0036	0.27	0.0001	0.16	0.0001			1.86	0.0025	1.99	0.0048
	2	9.146	7	7	4	1	2	3	7	9	3	2	3	0.000	0.00000	3	5	2	6
	P2	19.23	100	0.14	0.0000	0.36	0.0001	1.18	0.0014	0.27	0.0005	0.16	0.0000			1.87	0.0029	2.02	0.0026
	5	1	3	1	2	5	4	7	4	5	4	9	3	0.000	0.00000	2	2	3	2
	P2	31.25		0.13	0.0000	0.40	0.0008	1.30	0.0013	0.29	0.0012	0.17	0.0000			1.85	0.0020	2.02	0.0030
	6	3	947	3	6	3	9	7	3	6	9	3	9	0.000	0.00000	1	1	1	9

Figure S1. Maps showing the Catimbau National Park (white star) and Caatinga dry forest (black line) in Brazil (A), and the distribution of plots (circles) across gradients of disturbance (global multi-metric disturbance index – GMDI) and aridity (water deficit) intensity inside the Park (B).

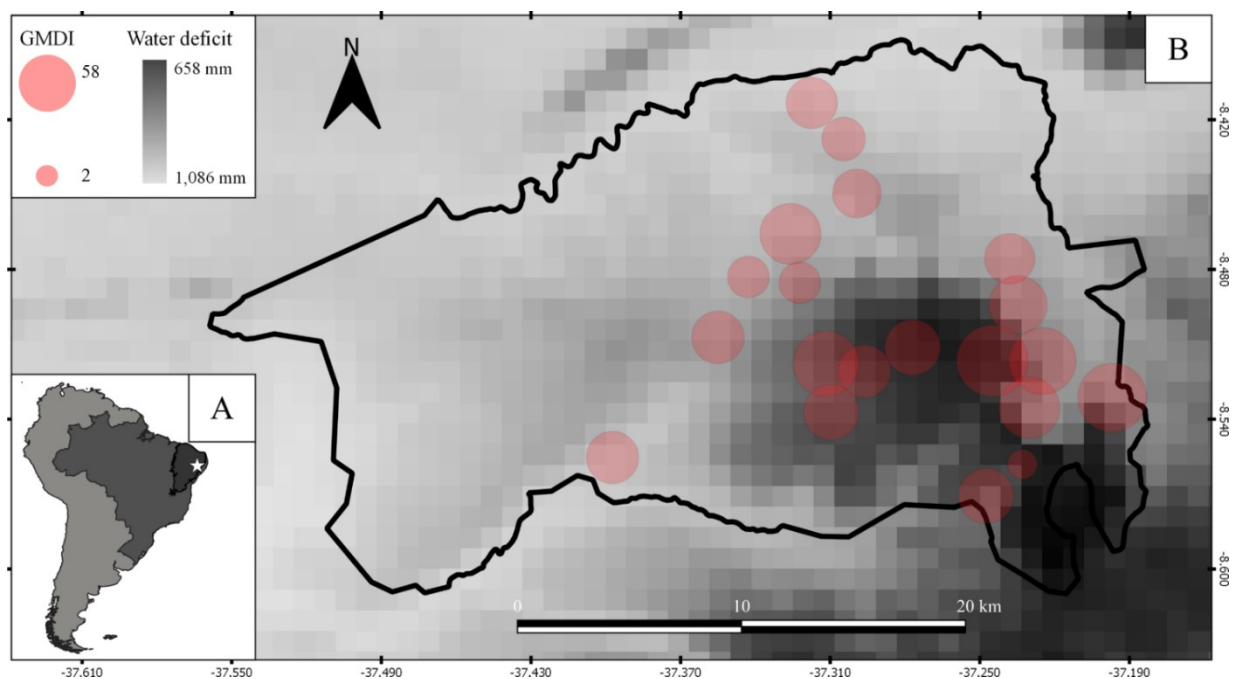


Figure S2. Plot distribution of (A) *Dinoponera quadriceps*, (B) *Ectatomma muticum*, (C) *Dorymyrmex goeldii* and (D) *Pheidole radoskowskii* along the disturbance and aridity gradients in the Catimbau National Park, north-eastern Brazil.

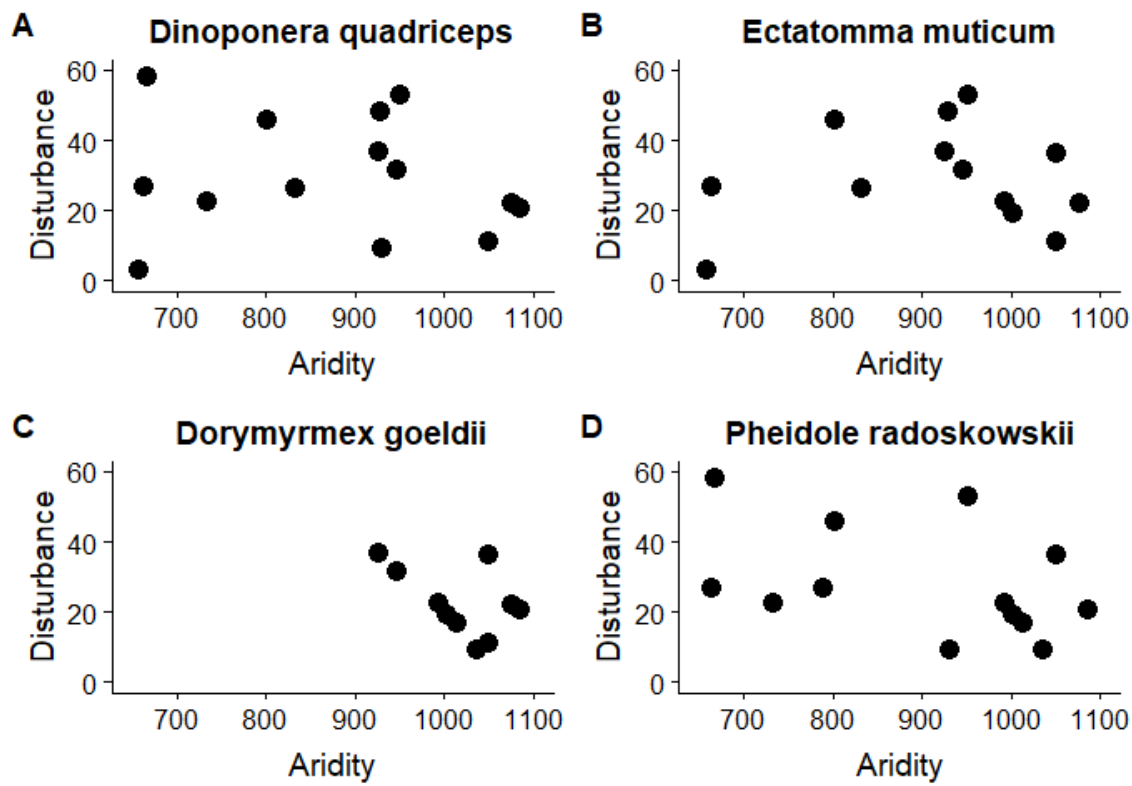


Figure S3. Corplot showing Pearson correlation coefficients of all traits across all individuals of (A) *Dinoponera quadriceps*, (B) *Ectatomma muticum*, (C) *Dorymyrmex goeldii* and (D) *Pheidole radoskowskii* measured along the aridity and disturbance gradients in the Catimbau National Park, north-eastern Brazil. Positive correlations are displayed in blue and negative correlations in red, color intensity is proportional to the correlation coefficients. Crosses represent no significant correlations (p-value > 0.05). CL = ratio of clypeus length to mesosoma length, ML = ratio of mandible length to mesosoma length, WL = mesosoma or Weber's length, IOW = ratio of inter-ocular width to mesosoma length, EL = ratio of eye length to mesosoma length, P = count of hairs crossing mesosoma profile, LEG = ratio of leg length (combined length of femur and tibia) to mesosoma length and AL = ratio of antenna (combined funicle and scape) length to mesosoma length.

